

Quantum Field Theory

Lecture Notes

Quantization, Fock space, propagators, scattering, and renormalization.



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Preface: What These Notes Are For

Remark

These notes are designed as a continuation of the Classical Field Theory I–II sequence. They do not try to re-teach variational calculus, Lorentz covariance, Noether currents, gauge redundancy, or the classical equations of fundamental fields from the beginning. Instead, they start where the classical notes stop: with fields as dynamical systems ready to be quantized.

Quantum Field Theory After Classical Field Theory

Classical field theory teaches us to describe nature by local fields, actions, symmetries, conservation laws, and equations of motion. Quantum field theory keeps this architecture, but changes the meaning of the basic objects. A classical field configuration is no longer the final description of a physical state. It becomes an ingredient in a quantum theory whose observable content is encoded in states, operators, correlation functions, transition amplitudes, and probabilities.

The guiding idea of the first part of these notes is simple: a free classical field decomposes into normal modes, and each normal mode behaves like a harmonic oscillator. Quantizing the field therefore means quantizing infinitely many oscillators at once. The excitations of these oscillators are interpreted as particles.

What Will Not Be Repeated

The notes assume familiarity with the following ideas from classical field theory:

- the action principle and Euler–Lagrange equations for fields,
- canonical momenta and Hamiltonian density,
- Lorentz covariance and relativistic notation,
- real and complex scalar fields,
- electromagnetic and Yang–Mills gauge fields at the classical level,
- Noether’s theorem and conserved currents,
- Green’s functions as solutions of linear differential equations.

Whenever one of these topics is used in a quantum calculation, the notes recall just enough of the classical structure to make the calculation readable. The aim is not repetition, but conversion: classical structures are translated into quantum structures.

What Is New in QFT

The genuinely new concepts are the vacuum, Fock space, field operators, propagators, Wick contractions, Feynman diagrams, scattering amplitudes, loop corrections, regularization, renormalization, and scale dependence. These ideas are not decorative additions to classical field theory. They are the mechanisms that make quantum field theory predictive.

A central conceptual warning will appear repeatedly: a Feynman diagram is not a literal picture of tiny particles following tiny trajectories. It is a compact bookkeeping device for terms in a perturbative expansion of correlation functions or scattering amplitudes.

How to Read These Notes

A useful rhythm for reading a QFT calculation is:

classical field $\longrightarrow$ mode expansion $\longrightarrow$ operators $\longrightarrow$ states $\longrightarrow$ correlators $\longrightarrow$ amplitudes.

At first this may look like a change of notation. It is not. Each arrow changes what is meant by a physical prediction. The purpose of the notes is to make those changes explicit and technically usable.

Chapter 1

From Classical Normal Modes to Quantum Oscillators

Remark

A free field is the continuum limit of a system with many coupled degrees of freedom. Its normal modes behave like independent harmonic oscillators. The first idea of quantum field theory is to quantize these oscillators and to interpret their excitations as field quanta.

1.1 The Purpose of the First Step

The transition from classical field theory to quantum field theory can look too large if one begins immediately with operator-valued distributions, Fock spaces, and relativistic commutators. This chapter takes a slower route. It explains why the harmonic oscillator algebra appears at the beginning of quantum field theory.

The guiding chain of ideas is

classical field $\longrightarrow$ normal modes $\longrightarrow$ classical oscillators $\longrightarrow$ quantum oscillators $\longrightarrow$ field quanta.

The goal is not yet to construct the full canonical field algebra. That will be done in the next chapter. Here we only identify the mechanical structure hidden inside a free field.

1.2 A Finite System of Coupled Oscillators

Consider a finite chain of N identical masses connected by springs. Let $q_j(t)$ be the displacement of the j -th mass from equilibrium. A typical quadratic Lagrangian has the form

$$L = \sum_{j=1}^N \frac{m}{2} \dot{q}_j^2 - \sum_{j=1}^N \frac{\kappa}{2} (q_{j+1} - q_j)^2,$$

with boundary conditions chosen so that the notation is meaningful. The coordinates q_j are coupled because the potential energy depends on differences between neighbouring displacements.

The normal-mode method replaces the original coordinates by new coordinates $Q_r(t)$,

$$q_j(t) = \sum_r Q_r(t) u_r(j),$$

where $u_r(j)$ are spatial mode functions. With an appropriate normalization, the Lagrangian becomes diagonal:

$$L = \sum_r \left(\frac{1}{2} \dot{Q}_r^2 - \frac{1}{2} \omega_r^2 Q_r^2 \right).$$

Thus the coupled system is equivalent to a collection of independent harmonic oscillators.

Example

The original variables q_j describe local displacements of the masses. The normal coordinates Q_r describe collective oscillation patterns of the whole chain. Quantization is easiest in the normal coordinates because the modes are independent.

1.3 The Field as a Continuum Limit

A field is the continuum analogue of a system with one degree of freedom at each spatial point. In a one-dimensional chain with lattice spacing a , the site position is $x_j = ja$, and one may think schematically of

$$q_j(t) \approx \phi(t, x_j).$$

As $a \rightarrow 0$, sums over sites become spatial integrals, and discrete normal modes become Fourier modes.

For a real scalar field in a finite box of volume V , periodic boundary conditions make the allowed momenta discrete. A schematic Fourier expansion is

$$\phi(t, \mathbf{x}) = \frac{1}{\sqrt{V}} \sum_{\mathbf{k}} q_{\mathbf{k}}(t) e^{i\mathbf{k} \cdot \mathbf{x}}.$$

For a real field the complex coefficients are not all independent; one must impose the appropriate reality condition. Equivalently, one may use real sine and cosine modes. The important point is independent of this choice: after the free field is decomposed into normal modes, each independent mode behaves like a classical oscillator.

1.4 Each Free Mode Is a Harmonic Oscillator

The free real scalar field has Lagrangian

$$L = \int d^3x \left(\frac{1}{2} \dot{\phi}^2 - \frac{1}{2} (\nabla \phi)^2 - \frac{1}{2} m^2 \phi^2 \right).$$

Its equation of motion is the Klein–Gordon equation,

$$(\partial_t^2 - \nabla^2 + m^2)\phi(t, \mathbf{x}) = 0.$$

After the free field is diagonalized into normal modes, the Lagrangian takes the schematic oscillator form

$$L = \sum_{\mathbf{k}} \left(\frac{1}{2} \dot{q}_{\mathbf{k}}^2 - \frac{1}{2} \omega_{\mathbf{k}}^2 q_{\mathbf{k}}^2 \right),$$

where

$$\omega_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + m^2}.$$

The canonical momentum of a mode is

$$p_{\mathbf{k}} = \dot{q}_{\mathbf{k}},$$

and the Hamiltonian is

$$H = \sum_{\mathbf{k}} H_{\mathbf{k}}, \quad H_{\mathbf{k}} = \frac{1}{2} p_{\mathbf{k}}^2 + \frac{1}{2} \omega_{\mathbf{k}}^2 q_{\mathbf{k}}^2.$$

A free scalar field is therefore a collection of independent harmonic oscillators, one for each independent normal mode.

Remark

Interactions destroy this simple diagonal form. Perturbative QFT begins by quantizing the free oscillators exactly and then treating interactions as corrections to the free theory.

1.5 Quantizing One Harmonic Oscillator

Before quantizing all field modes, recall the single harmonic oscillator. The classical variables q and p become operators $\hat{q}$ and $\hat{p}$ satisfying

$$[\hat{q}, \hat{p}] = i.$$

The Hamiltonian is

$$\hat{H} = \frac{1}{2}\hat{p}^2 + \frac{1}{2}\omega^2\hat{q}^2.$$

Introduce the annihilation and creation operators

$$\hat{a} = \sqrt{\frac{\omega}{2}}\hat{q} + \frac{i}{\sqrt{2\omega}}\hat{p}, \quad \hat{a}^\dagger = \sqrt{\frac{\omega}{2}}\hat{q} - \frac{i}{\sqrt{2\omega}}\hat{p}.$$

Then

$$[\hat{a}, \hat{a}^\dagger] = 1,$$

and

$$\hat{H} = \omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right).$$

The operator $\hat{a}^\dagger \hat{a}$ counts oscillator excitations.

1.6 Number States and the Oscillator Vacuum

The oscillator vacuum is defined by

$$\hat{a}|0\rangle = 0.$$

The excited states are

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle, \quad n = 0, 1, 2, \dots$$

If

$$\hat{N} = \hat{a}^\dagger \hat{a},$$

then

$$\hat{N}|n\rangle = n|n\rangle, \quad \hat{H}|n\rangle = \omega \left(n + \frac{1}{2} \right) |n\rangle.$$

The vacuum is not a state of zero energy. It has the zero-point energy $\omega/2$. This fact becomes more delicate for fields because a field has infinitely many oscillator modes.

1.7 From One Oscillator to Many Field Modes

For a free field, every independent mode is quantized as an oscillator. Thus one introduces a pair of operators for each mode,

$$\hat{a}_{\mathbf{k}}, \quad \hat{a}_{\mathbf{k}}^\dagger.$$

In a finite box their algebra is

$$[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{p}}^\dagger] = \delta_{\mathbf{k}, \mathbf{p}}, \quad [\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{p}}] = 0, \quad [\hat{a}_{\mathbf{k}}^\dagger, \hat{a}_{\mathbf{p}}^\dagger] = 0.$$

The free-field vacuum is the state annihilated by all annihilation operators,

$$\hat{a}_{\mathbf{k}}|0\rangle = 0 \quad \text{for all } \mathbf{k}.$$

A one-quantum state is obtained by applying one creation operator:

$$|\mathbf{k}\rangle = \hat{a}_{\mathbf{k}}^\dagger|0\rangle.$$

A many-quantum state is obtained by applying several creation operators.

Definition

The *Fock space* of the free field is the Hilbert space generated from the vacuum by applying creation operators for all modes. It naturally contains states with zero, one, two, and more field quanta.

1.8 Field Quanta as Mode Excitations

For the free theory, the Hamiltonian has the schematic form

$$\hat{H} = \sum_{\mathbf{k}} \omega_{\mathbf{k}} \left(\hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} + \frac{1}{2} \right).$$

The operator $\hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}}$ counts how many quanta occupy the mode $\mathbf{k}$. The state $\hat{a}_{\mathbf{k}}^\dagger|0\rangle$ has one quantum of energy $\omega_{\mathbf{k}}$ above the free vacuum and momentum $\mathbf{k}$. In the free theory, this is what one means by a one-particle state.

This interpretation is precise for a free field. In interacting QFT, particles can be dressed by interactions, unstable particles appear as resonances, and the vacuum itself can be nontrivial. Nevertheless, the free Fock-space picture is the starting point for perturbation theory and scattering.

1.9 What Is Still Missing?

The oscillator construction explains why creation and annihilation operators appear, but it is not yet the full canonical quantization of a field. We have not yet imposed commutation relations directly on $\hat{\phi}(t, \mathbf{x})$ and its conjugate momentum $\hat{\pi}(t, \mathbf{x})$. We have also not fixed the continuum normalization of the mode expansion, nor have we discussed fields as operator-valued distributions.

These are the tasks of the next chapter. Chapter 2 turns the oscillator picture into the canonical quantization of the real scalar field.

Summary

A free classical field can be diagonalized into independent normal modes. Each mode is a harmonic oscillator. Quantizing those oscillators gives creation and annihilation operators. Acting with creation operators on the vacuum builds Fock space, and the resulting mode excitations are interpreted as free particles. This oscillator picture is the conceptual bridge to canonical field quantization.

1.10 Chapter Checklist

After reading this chapter, one should be able to:

- explain why a free field decomposes into normal modes;
- explain why each independent mode behaves as a harmonic oscillator;
- quantize a single harmonic oscillator using $\hat{a}$ and $\hat{a}^\dagger$;
- define the oscillator vacuum and number states;
- describe the free-field vacuum and one-particle states;
- explain why free particles are mode excitations;
- state what remains to be done in canonical field quantization.

Chapter 2

Canonical Quantization of the Real Scalar Field

Remark

Chapter 1 explained why a free field should be thought of as a collection of oscillators. This chapter makes that statement precise. We impose canonical commutation relations on the field and its conjugate momentum, derive the standard mode expansion, and recover the Fock-space interpretation from the field algebra itself.

2.1 From the Oscillator Picture to Field Operators

The previous chapter started from normal modes. It argued that a free scalar field is a collection of harmonic oscillators labelled by momentum. That is the right intuition, but it is not yet the full canonical quantization of a field.

The full canonical construction begins before the normal-mode expansion. It starts with the classical field $\phi(t, \mathbf{x})$, its conjugate momentum $\pi(t, \mathbf{x})$, and their Poisson brackets. Quantization then promotes ϕ and π to operators and replaces the Poisson brackets by commutators.

The logic of this chapter is therefore

classical field variables $\longrightarrow$ equal-time commutators $\longrightarrow$ operator mode expansion $\longrightarrow$ creation and annihilation

This is the formal version of the oscillator picture developed in Chapter 1.

2.2 Classical Starting Point

We use units $c = \hbar = 1$ and the metric convention $(+, -, -, -)$. The free real scalar field is described by the Lagrangian density

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 = \frac{1}{2} \dot{\phi}^2 - \frac{1}{2} (\nabla \phi)^2 - \frac{1}{2} m^2 \phi^2.$$

The conjugate momentum is

$$\pi(t, \mathbf{x}) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}(t, \mathbf{x})} = \dot{\phi}(t, \mathbf{x}).$$

The Euler–Lagrange equation is the Klein–Gordon equation,

$$(\square + m^2)\phi(x) = 0,$$

where $x = (t, \mathbf{x})$. In components this reads

$$(\partial_t^2 - \nabla^2 + m^2)\phi(t, \mathbf{x}) = 0.$$

The classical Hamiltonian density is

$$\mathcal{H} = \frac{1}{2} \pi^2 + \frac{1}{2} (\nabla \phi)^2 + \frac{1}{2} m^2 \phi^2,$$

so the Hamiltonian is

$$H = \int d^3x \left(\frac{1}{2} \pi^2 + \frac{1}{2} (\nabla \phi)^2 + \frac{1}{2} m^2 \phi^2 \right).$$

This is the continuum field-theoretic analogue of the oscillator Hamiltonian $p^2/2 + \omega^2 q^2/2$.

2.3 Equal-Time Commutation Relations

The classical canonical Poisson bracket is

$$\{\phi(t, \mathbf{x}), \pi(t, \mathbf{y})\}_{\text{PB}} = \delta^{(3)}(\mathbf{x} - \mathbf{y}).$$

Canonical quantization promotes ϕ and π to operators $\hat{\phi}$ and $\hat{\pi}$, and imposes

$$[\hat{\phi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{y})] = i\delta^{(3)}(\mathbf{x} - \mathbf{y}).$$

The remaining equal-time commutators vanish:

$$[\hat{\phi}(t, \mathbf{x}), \hat{\phi}(t, \mathbf{y})] = 0, \quad [\hat{\pi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{y})] = 0.$$

These relations are the field version of

$$[\hat{q}, \hat{p}] = i.$$

Instead of one coordinate and one momentum, there is one canonical pair at every point of space.

Definition

The *equal-time canonical commutation relations* are the basic operator relations between $\hat{\phi}$ and $\hat{\pi}$ at a fixed time. For bosonic fields they are commutators; for fermionic fields they will be replaced by anticommutators.

2.4 Finite Box and Continuum Normalization

Chapter 1 used a finite box to keep the momentum labels discrete. In a finite box, one writes sums over momenta and Kronecker deltas. In infinite volume, one uses integrals and Dirac deltas. The replacement rules are schematically

$$\frac{1}{V} \sum_{\mathbf{k}} \longrightarrow \int \frac{d^3k}{(2\pi)^3}, \quad V\delta_{\mathbf{k},\mathbf{p}} \longrightarrow (2\pi)^3\delta^{(3)}(\mathbf{k} - \mathbf{p}).$$

The continuum formulas below use the second convention. They look slightly more complicated than finite-box formulas, but they are the standard relativistic normalization used in scattering theory.

2.5 Mode Expansion of the Field Operator

The quantum field operator is expanded in positive- and negative-frequency solutions of the Klein–Gordon equation:

$$\hat{\phi}(t, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{k}}}} \left(\hat{a}_{\mathbf{k}} e^{-i\omega_{\mathbf{k}}t + i\mathbf{k}\cdot\mathbf{x}} + \hat{a}_{\mathbf{k}}^\dagger e^{i\omega_{\mathbf{k}}t - i\mathbf{k}\cdot\mathbf{x}} \right),$$

where

$$\omega_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + m^2}.$$

Since the field is real, $\hat{\phi}$ is Hermitian:

$$\hat{\phi}(t, \mathbf{x})^\dagger = \hat{\phi}(t, \mathbf{x}).$$

The appearance of both $\hat{a}_{\mathbf{k}}$ and $\hat{a}_{\mathbf{k}}^\dagger$ is the operator version of the reality condition on a classical real field.

The conjugate momentum operator is

$$\hat{\pi}(t, \mathbf{x}) = \partial_t \hat{\phi}(t, \mathbf{x}),$$

therefore

$$\hat{\pi}(t, \mathbf{x}) = -i \int \frac{d^3k}{(2\pi)^3} \sqrt{\frac{\omega_{\mathbf{k}}}{2}} \left(\hat{a}_{\mathbf{k}} e^{-i\omega_{\mathbf{k}}t + i\mathbf{k}\cdot\mathbf{x}} - \hat{a}_{\mathbf{k}}^\dagger e^{i\omega_{\mathbf{k}}t - i\mathbf{k}\cdot\mathbf{x}} \right).$$

The normalization factors are chosen so that the equal-time field commutators are satisfied.

2.6 Ladder-Operator Algebra

The canonical field commutators imply

$$[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{p}}^\dagger] = (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{p}),$$

with

$$[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{p}}] = 0, \quad [\hat{a}_{\mathbf{k}}^\dagger, \hat{a}_{\mathbf{p}}^\dagger] = 0.$$

Conversely, these ladder-operator relations imply the equal-time commutators of $\hat{\phi}$ and $\hat{\pi}$. This is the precise connection between the canonical field algebra and the oscillator algebra of Chapter 1.

The operator $\hat{a}_{\mathbf{k}}^\dagger$ creates a quantum in the momentum mode $\mathbf{k}$, while $\hat{a}_{\mathbf{k}}$ annihilates such a quantum. For the real scalar field there is only one kind of neutral scalar quantum.

2.7 Fock Space Revisited

The free-field vacuum is defined by

$$\hat{a}_{\mathbf{k}}|0\rangle = 0 \quad \text{for every } \mathbf{k}.$$

A one-particle momentum eigenstate is

$$|\mathbf{k}\rangle = \hat{a}_{\mathbf{k}}^\dagger|0\rangle.$$

A two-particle state is

$$|\mathbf{k}_1, \mathbf{k}_2\rangle = \hat{a}_{\mathbf{k}_1}^\dagger \hat{a}_{\mathbf{k}_2}^\dagger|0\rangle,$$

up to normalization. Since the creation operators commute, the scalar quanta are bosons.

A realistic one-particle state is usually a wave packet rather than a single momentum eigenstate:

$$|f\rangle = \int \frac{d^3k}{(2\pi)^3} f(\mathbf{k}) \hat{a}_{\mathbf{k}}^\dagger|0\rangle.$$

The function $f(\mathbf{k})$ describes the momentum profile of the state. This is the first place where the particle idea becomes more refined than the simple phrase “one quantum of one mode.”

2.8 Hamiltonian and Momentum Operators

The quantum Hamiltonian is obtained from the classical Hamiltonian by replacing ϕ and π by field operators:

$$\hat{H} = \int d^3x \left(\frac{1}{2} \hat{\pi}^2 + \frac{1}{2} (\nabla \hat{\phi})^2 + \frac{1}{2} m^2 \hat{\phi}^2 \right).$$

Substituting the mode expansion gives the formal expression

$$\hat{H} = \int \frac{d^3k}{(2\pi)^3} \omega_{\mathbf{k}} \left(\hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} + \frac{1}{2} (2\pi)^3 \delta^{(3)}(0) \right).$$

The first term counts excitations; the second term is the continuum notation for the sum of oscillator zero-point energies.

The spatial momentum operator is

$$\hat{\mathbf{P}} = \int d^3x \hat{\pi} \nabla \hat{\phi}.$$

With the same ordering convention, it becomes

$$\hat{\mathbf{P}} = \int \frac{d^3k}{(2\pi)^3} \mathbf{k} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}}.$$

Thus

$$\hat{H} \hat{a}_{\mathbf{k}}^\dagger |0\rangle = \omega_{\mathbf{k}} \hat{a}_{\mathbf{k}}^\dagger |0\rangle \quad \text{above the vacuum energy,}$$

and

$$\hat{\mathbf{P}} \hat{a}_{\mathbf{k}}^\dagger |0\rangle = \mathbf{k} \hat{a}_{\mathbf{k}}^\dagger |0\rangle.$$

The dispersion relation is

$$\omega_{\mathbf{k}}^2 = \mathbf{k}^2 + m^2.$$

This is why m is interpreted as the mass of the scalar quantum.

2.9 Normal Ordering

The vacuum-energy term in $\hat{H}$ is infinite in the continuum limit. For the free theory, it is common to define the normal-ordered Hamiltonian by moving all creation operators to the left of all annihilation operators:

$$: \hat{a}_{\mathbf{k}} \hat{a}_{\mathbf{p}}^\dagger := \hat{a}_{\mathbf{p}}^\dagger \hat{a}_{\mathbf{k}}.$$

Then

$$: \hat{H} := \int \frac{d^3k}{(2\pi)^3} \omega_{\mathbf{k}} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}},$$

and

$$\langle 0 | : \hat{H} : | 0 \rangle = 0.$$

Normal ordering removes the constant free vacuum energy from this Hamiltonian. It does not remove all divergences of interacting quantum field theory. It is a useful convention at the free-field level, not a substitute for renormalization.

2.10 Zero-Point Energy

In a finite box, the vacuum energy of the free scalar field is

$$E_0 = \frac{1}{2} \sum_{\mathbf{k}} \omega_{\mathbf{k}}.$$

In infinite-volume notation this becomes

$$E_0 = \frac{V}{2} \int \frac{d^3k}{(2\pi)^3} \omega_{\mathbf{k}}.$$

The integral diverges at large momentum. The divergence is a reminder that a field has infinitely many short-distance degrees of freedom.

For many non-gravitational scattering problems, only energy differences above the free vacuum matter, and normal ordering is sufficient. In contexts where vacuum energy gravitates, or where the vacuum itself changes, the issue becomes more subtle. The present course will return to this point when discussing renormalization and vacuum structure.

2.11 Fields as Operator-Valued Distributions

The notation $\hat{\phi}(x)$ suggests an operator at a point. This notation is useful, but local quantum fields are more accurately understood as operator-valued distributions. They become well-defined only after smearing with test functions.

At fixed time, a smeared field is

$$\hat{\phi}(f) = \int d^3x f(\mathbf{x}) \hat{\phi}(t, \mathbf{x}).$$

Products such as $\hat{\phi}(x)^2$ at the same spacetime point are generally singular and require a prescription. This is one of the structural reasons why renormalization appears in interacting field theory.

This point also clarifies the role of the mode expansion. The formula for $\hat{\phi}(t, \mathbf{x})$ is a compact way to encode how the field acts after it has been smeared. It should not be treated as an ordinary function with operator coefficients.

2.12 Microcausality

Relativistic quantum field theory must respect the causal structure of spacetime. For a scalar field, the basic local causality condition is

$$[\hat{\phi}(x), \hat{\phi}(y)] = 0 \quad \text{if} \quad (x - y)^2 < 0.$$

This condition is called *microcausality*. It says that field observables associated with spacelike separated regions commute.

For the free scalar field one can write

$$[\hat{\phi}(x), \hat{\phi}(y)] = i\Delta(x - y),$$

where Δ is the Pauli–Jordan commutator function. This function vanishes outside the light cone. The equal-time relation

$$[\hat{\phi}(t, \mathbf{x}), \hat{\phi}(t, \mathbf{y})] = 0$$

is therefore only the simplest special case of the full spacetime causality condition.

Summary

Canonical quantization begins with $\hat{\phi}$ and $\hat{\pi}$, not directly with particles. The equal-time commutation relations determine the ladder operator algebra in the mode expansion. The free-field vacuum and Fock space are then recovered from that algebra, matching the oscillator picture of Chapter 1. The new lessons are the precise continuum normalization, the Hamiltonian and momentum operators, normal ordering, the distributional nature of fields, and microcausality.

2.13 Chapter Checklist

After reading this chapter, one should be able to:

- write the classical Lagrangian and Hamiltonian of the free real scalar field;
- derive the conjugate momentum $\pi = \dot{\phi}$;
- state the equal-time canonical commutation relations;

- write the mode expansions of $\hat{\phi}$ and $\hat{\pi}$;
- derive the ladder-operator commutators from the canonical field algebra;
- construct the vacuum, one-particle states, wave packets, and many-particle states;
- express $\hat{H}$ and $\hat{\mathbf{P}}$ in terms of ladder operators;
- explain what normal ordering removes and what it does not remove;
- explain why fields are operator-valued distributions;
- state the microcausality condition for a scalar field.

Chapter 3

The Complex Scalar Field and Antiparticles

Remark

The real scalar field has one neutral quantum for each momentum mode. The complex scalar field is the next step: the field is no longer Hermitian, and its quantization produces two independent kinds of quanta. They are naturally interpreted as particles and antiparticles.

3.1 From a Real Field to a Complex Field

Chapter 2 quantized a real scalar field. The reality condition $\hat{\phi}^\dagger = \hat{\phi}$ forced the same field operator to contain both annihilation and creation operators. A complex scalar field is different. The field ϕ and its Hermitian conjugate $\phi^\dagger$ are independent classical variables before quantization.

One may think of a complex scalar field as two real scalar fields packaged into one object:

$$\phi = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2), \quad \phi^\dagger = \frac{1}{\sqrt{2}}(\phi_1 - i\phi_2).$$

The important new feature is that the theory has a global phase symmetry. The field may be rotated by a constant phase,

$$\phi \longrightarrow e^{-i\alpha}\phi, \quad \phi^\dagger \longrightarrow e^{i\alpha}\phi^\dagger.$$

By Noether's theorem this symmetry gives a conserved charge. In the quantum theory, that charge will distinguish particles from antiparticles.

3.2 Classical Lagrangian and Canonical Momenta

The free complex scalar field is described by

$$\mathcal{L} = \partial_\mu \phi^\dagger \partial^\mu \phi - m^2 \phi^\dagger \phi.$$

In terms of time and space derivatives,

$$\mathcal{L} = \dot{\phi}^\dagger \dot{\phi} - \nabla \phi^\dagger \cdot \nabla \phi - m^2 \phi^\dagger \phi.$$

The conjugate momenta are

$$\pi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \dot{\phi}^\dagger, \quad \pi^\dagger = \frac{\partial \mathcal{L}}{\partial \dot{\phi}^\dagger} = \dot{\phi}.$$

Thus π is conjugate to ϕ , while $\pi^\dagger$ is conjugate to $\phi^\dagger$.

The equations of motion are

$$(\square + m^2)\phi = 0, \quad (\square + m^2)\phi^\dagger = 0.$$

Both components satisfy the Klein–Gordon equation, but the quantum interpretation will be richer than in the real case.

3.3 Canonical Commutation Relations

Canonical quantization promotes $\phi, \phi^\dagger, \pi, \pi^\dagger$ to operators. The nonzero equal-time commutators are

$$[\hat{\phi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{y})] = i\delta^{(3)}(\mathbf{x} - \mathbf{y}),$$

and

$$[\hat{\phi}^\dagger(t, \mathbf{x}), \hat{\pi}^\dagger(t, \mathbf{y})] = i\delta^{(3)}(\mathbf{x} - \mathbf{y}).$$

All other equal-time commutators vanish. In particular,

$$[\hat{\phi}(t, \mathbf{x}), \hat{\pi}^\dagger(t, \mathbf{y})] = 0, \quad [\hat{\phi}^\dagger(t, \mathbf{x}), \hat{\pi}(t, \mathbf{y})] = 0.$$

This is the canonical field algebra for a charged bosonic scalar field.

3.4 Mode Expansion

The complex field is not Hermitian, so its mode expansion cannot be the same as the real scalar expansion. The standard continuum expansion is

$$\hat{\phi}(t, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{k}}}} \left(\hat{a}_{\mathbf{k}} e^{-i\omega_{\mathbf{k}}t + i\mathbf{k}\cdot\mathbf{x}} + \hat{b}_{\mathbf{k}}^\dagger e^{i\omega_{\mathbf{k}}t - i\mathbf{k}\cdot\mathbf{x}} \right),$$

where

$$\omega_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + m^2}.$$

The Hermitian conjugate field is

$$\hat{\phi}^\dagger(t, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{k}}}} \left(\hat{a}_{\mathbf{k}}^\dagger e^{i\omega_{\mathbf{k}}t - i\mathbf{k}\cdot\mathbf{x}} + \hat{b}_{\mathbf{k}} e^{-i\omega_{\mathbf{k}}t + i\mathbf{k}\cdot\mathbf{x}} \right).$$

There are now two independent sets of ladder operators: the a -operators and the b -operators. This is the algebraic origin of antiparticles in the complex scalar theory.

3.5 Two Independent Oscillator Families

The nonzero ladder-operator commutators are

$$[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{p}}^\dagger] = (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{p}),$$

and

$$[\hat{b}_{\mathbf{k}}, \hat{b}_{\mathbf{p}}^\dagger] = (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{p}).$$

The two families commute with each other:

$$[\hat{a}_{\mathbf{k}}, \hat{b}_{\mathbf{p}}] = 0, \quad [\hat{a}_{\mathbf{k}}, \hat{b}_{\mathbf{p}}^\dagger] = 0,$$

with similar relations for their Hermitian conjugates.

The vacuum is defined by

$$\hat{a}_{\mathbf{k}}|0\rangle = 0, \quad \hat{b}_{\mathbf{k}}|0\rangle = 0 \quad \text{for every } \mathbf{k}.$$

There are two basic one-particle states:

$$|\mathbf{k}; a\rangle = \hat{a}_{\mathbf{k}}^\dagger|0\rangle, \quad |\mathbf{k}; b\rangle = \hat{b}_{\mathbf{k}}^\dagger|0\rangle.$$

They have the same mass and the same energy-momentum relation, but opposite charges.

3.6 Hamiltonian and Momentum

After normal ordering, the Hamiltonian of the free complex scalar field is

$$:\hat{H}:=\int\frac{d^3k}{(2\pi)^3}\omega_{\mathbf{k}}\left(\hat{a}_{\mathbf{k}}^\dagger\hat{a}_{\mathbf{k}}+\hat{b}_{\mathbf{k}}^\dagger\hat{b}_{\mathbf{k}}\right).$$

The momentum operator is

$$:\hat{\mathbf{P}}:=\int\frac{d^3k}{(2\pi)^3}\mathbf{k}\left(\hat{a}_{\mathbf{k}}^\dagger\hat{a}_{\mathbf{k}}+\hat{b}_{\mathbf{k}}^\dagger\hat{b}_{\mathbf{k}}\right).$$

Both kinds of quanta carry positive energy. The difference between them is not energy or mass, but charge.

3.7 The Charge Operator

The global phase symmetry gives a conserved current. With the convention $\phi \rightarrow e^{-i\alpha}\phi$, the corresponding quantum charge may be written, after normal ordering, as

$$\hat{Q}=\int\frac{d^3k}{(2\pi)^3}\left(\hat{a}_{\mathbf{k}}^\dagger\hat{a}_{\mathbf{k}}-\hat{b}_{\mathbf{k}}^\dagger\hat{b}_{\mathbf{k}}\right).$$

This convention gives

$$\hat{Q}\hat{a}_{\mathbf{k}}^\dagger|0\rangle=+\hat{a}_{\mathbf{k}}^\dagger|0\rangle,\quad\hat{Q}\hat{b}_{\mathbf{k}}^\dagger|0\rangle=-\hat{b}_{\mathbf{k}}^\dagger|0\rangle.$$

Thus the a -quanta have charge $+1$, while the b -quanta have charge -1 . They are particle and antiparticle excitations of the same complex field.

The charge operator also generates the phase transformation of the field:

$$[\hat{Q},\hat{\phi}(x)]=-\hat{\phi}(x),\quad[\hat{Q},\hat{\phi}^\dagger(x)]=+\hat{\phi}^\dagger(x).$$

This is the operator statement that $\hat{\phi}$ carries charge -1 as an operator, even though $\hat{a}^\dagger$ creates a charge $+1$ state in the chosen convention.

3.8 Positive and Negative Frequency Revisited

In the real scalar field, the negative-frequency part was simply the Hermitian conjugate of the positive-frequency part of the same field. In the complex field, the negative-frequency part of $\hat{\phi}$ contains $\hat{b}^\dagger$, not $\hat{a}^\dagger$. This is the crucial difference.

The field $\hat{\phi}$ can annihilate an a -particle or create a b -antiparticle. The conjugate field $\hat{\phi}^\dagger$ can create an a -particle or annihilate a b -antiparticle. Symbolically,

$$\hat{\phi}\sim\hat{a}+\hat{b}^\dagger,\quad\hat{\phi}^\dagger\sim\hat{a}^\dagger+\hat{b}.$$

This is one of the cleanest ways to see why relativistic quantum fields contain antiparticles.

3.9 Why Antiparticles Appear Naturally

The appearance of antiparticles is not an optional addition. It follows from combining three requirements:

- relativistic positive-energy modes;
- canonical quantization;

- a complex field with a conserved charge.

The field must contain both positive- and negative-frequency solutions in order to be local and to satisfy the canonical commutation relations. For a charged field, those two frequency sectors are associated with two independent oscillator families. The quanta created by the second family have the opposite charge and are interpreted as antiparticles.

3.10 Complex Scalar Propagator

The natural two-point function of the complex scalar field is

$$D_F(x - y) = \langle 0 | T \hat{\phi}(x) \hat{\phi}^\dagger(y) | 0 \rangle.$$

It has the same momentum-space denominator as the real scalar propagator:

$$D_F(x - y) = \int \frac{d^4 k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} e^{-ik \cdot (x-y)}.$$

The difference is in the charge flow. The propagator connects ϕ with $\phi^\dagger$, because charge must be conserved. Correlators such as $\langle 0 | T \hat{\phi}(x) \hat{\phi}(y) | 0 \rangle$ vanish in the free theory when the vacuum has zero charge.

3.11 Physical Interpretation of the Fock Space

The Fock space of the complex scalar field contains particles, antiparticles, and arbitrary multiparticle states. Examples are

$$\hat{a}_{\mathbf{k}}^\dagger |0\rangle, \quad \hat{b}_{\mathbf{p}}^\dagger |0\rangle, \quad \hat{a}_{\mathbf{k}}^\dagger \hat{b}_{\mathbf{p}}^\dagger |0\rangle.$$

The last state contains one particle and one antiparticle. Its total charge is zero, but it is not the vacuum. Charge conservation restricts which transitions are possible once interactions are introduced.

Summary

A complex scalar field is equivalent to two real scalar degrees of freedom, but its global phase symmetry gives a conserved charge. Quantization produces two independent oscillator families. One creates particles and the other creates antiparticles, with equal mass and opposite charge. The field $\hat{\phi}$ contains an annihilation operator for particles and a creation operator for antiparticles, while $\hat{\phi}^\dagger$ does the reverse.

3.12 Chapter Checklist

After reading this chapter, one should be able to:

- explain why ϕ and $\phi^\dagger$ are independent field variables before quantization;
- write the canonical momenta of the complex scalar field;
- state the equal-time commutation relations;
- write the mode expansions of $\hat{\phi}$ and $\hat{\phi}^\dagger$;
- explain why two independent ladder-operator families appear;

- construct particle and antiparticle states;
- write the charge operator and interpret its eigenvalues;
- explain why antiparticles are required in a relativistic charged field;
- identify the charge structure of the complex scalar propagator.

Chapter 4

Quantization of the Dirac Field

Remark

Scalar fields describe spin-zero quanta and are quantized with commutators. The Dirac field describes spin-1/2 quanta. Its consistent quantization requires anticommutators, and its mode expansion naturally contains particle and antiparticle operators with spin labels.

4.1 Why Fermionic Quantization Is Different

Chapters 2 and 3 quantized bosonic scalar fields. The ladder operators obeyed commutation relations, and many-particle states were symmetric under exchange. The Dirac field is different because its quanta are fermions. Fermionic states must be antisymmetric under exchange, and no two identical fermions can occupy the same one-particle state.

This difference is implemented algebraically by replacing commutators with anticommutators. For operators $\hat{A}$ and $\hat{B}$, the anticommutator is

$$\{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}.$$

The quantized Dirac field will be built from creation and annihilation operators whose anticommutation relations enforce Fermi statistics.

There is also a second important difference. The Dirac equation is first order in time derivatives, unlike the Klein–Gordon equation. This changes the canonical structure and makes the Dirac field look less like a collection of ordinary oscillator coordinates. Nevertheless, the final Fock-space structure is parallel to the scalar case: there are annihilation operators, creation operators, a vacuum, particles, antiparticles, and number operators.

4.2 Classical Dirac Field

The Dirac field is a spinor field $\psi(x)$. Its Dirac adjoint is

$$\bar{\psi} = \psi^\dagger \gamma^0.$$

The free Dirac Lagrangian density is

$$\mathcal{L} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi.$$

The equation of motion is the Dirac equation,

$$(i\gamma^\mu \partial_\mu - m)\psi(x) = 0.$$

The adjoint field satisfies

$$\bar{\psi}(x)(i\overleftarrow{\partial}_\mu \gamma^\mu + m) = 0,$$

where the arrow indicates that the derivative acts to the left.

The matrices γ^μ obey the Clifford algebra

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}.$$

This algebra is what makes the Dirac equation compatible with the relativistic mass-shell relation $p^2 = m^2$.

4.3 Canonical Structure and Anticommutation Relations

Because the Lagrangian is first order in $\dot{\psi}$, the momentum conjugate to ψ is proportional to $\psi^\dagger$:

$$\pi_\psi = \frac{\partial \mathcal{L}}{\partial \dot{\psi}} = i\psi^\dagger.$$

The quantum theory is defined by equal-time anticommutation relations,

$$\{\hat{\psi}_\alpha(t, \mathbf{x}), \hat{\psi}_\beta^\dagger(t, \mathbf{y})\} = \delta_{\alpha\beta} \delta^{(3)}(\mathbf{x} - \mathbf{y}),$$

with

$$\{\hat{\psi}_\alpha(t, \mathbf{x}), \hat{\psi}_\beta(t, \mathbf{y})\} = 0, \quad \{\hat{\psi}_\alpha^\dagger(t, \mathbf{x}), \hat{\psi}_\beta^\dagger(t, \mathbf{y})\} = 0.$$

The spinor indices α, β label the components of the Dirac spinor.

Remark

The replacement of commutators by anticommutators is not a cosmetic change. It is what produces fermionic Fock space, positive-energy particle interpretation, and the Pauli exclusion principle.

4.4 Mode Expansion of the Dirac Field

The free Dirac equation has positive-energy spinor solutions $u_s(\mathbf{p})$ and negative-frequency spinor solutions $v_s(\mathbf{p})$, where s labels spin. A standard expansion is

$$\hat{\psi}(x) = \sum_s \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \left(\hat{b}_s(\mathbf{p}) u_s(\mathbf{p}) e^{-iE_{\mathbf{p}}t + i\mathbf{p}\cdot\mathbf{x}} + \hat{d}_s^\dagger(\mathbf{p}) v_s(\mathbf{p}) e^{iE_{\mathbf{p}}t - i\mathbf{p}\cdot\mathbf{x}} \right),$$

where

$$E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}.$$

The adjoint field is

$$\hat{\bar{\psi}}(x) = \sum_s \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \left(\hat{b}_s^\dagger(\mathbf{p}) \bar{u}_s(\mathbf{p}) e^{iE_{\mathbf{p}}t - i\mathbf{p}\cdot\mathbf{x}} + \hat{d}_s(\mathbf{p}) \bar{v}_s(\mathbf{p}) e^{-iE_{\mathbf{p}}t + i\mathbf{p}\cdot\mathbf{x}} \right).$$

The structure is analogous to the complex scalar field:

$$\hat{\psi} \sim \hat{b} + \hat{d}^\dagger, \quad \hat{\bar{\psi}} \sim \hat{b}^\dagger + \hat{d}.$$

The operators $\hat{b}^\dagger$ create particles, while $\hat{d}^\dagger$ create antiparticles.

4.5 Spinor Creation and Annihilation Operators

The nonzero anticommutators are

$$\{\hat{b}_s(\mathbf{p}), \hat{b}_r^\dagger(\mathbf{q})\} = (2\pi)^3 \delta_{sr} \delta^{(3)}(\mathbf{p} - \mathbf{q}),$$

and

$$\{\hat{d}_s(\mathbf{p}), \hat{d}_r^\dagger(\mathbf{q})\} = (2\pi)^3 \delta_{sr} \delta^{(3)}(\mathbf{p} - \mathbf{q}).$$

All other anticommutators vanish. In particular,

$$\{\hat{b}_s(\mathbf{p}), \hat{d}_r(\mathbf{q})\} = 0, \quad \{\hat{b}_s(\mathbf{p}), \hat{d}_r^\dagger(\mathbf{q})\} = 0.$$

The label s is the spin label. For a massive Dirac particle there are two spin states for the particle and two spin states for the antiparticle.

Because the creation operators anticommute, applying the same creation operator twice gives zero:

$$\left(\hat{b}_s^\dagger(\mathbf{p})\right)^2 = 0, \quad \left(\hat{d}_s^\dagger(\mathbf{p})\right)^2 = 0.$$

This is the algebraic form of the Pauli exclusion principle.

4.6 Particle and Antiparticle States

The vacuum is defined by

$$\hat{b}_s(\mathbf{p})|0\rangle = 0, \quad \hat{d}_s(\mathbf{p})|0\rangle = 0 \quad \text{for all } \mathbf{p}, s.$$

A one-particle state is

$$|\mathbf{p}, s; b\rangle = \hat{b}_s^\dagger(\mathbf{p})|0\rangle.$$

A one-antiparticle state is

$$|\mathbf{p}, s; d\rangle = \hat{d}_s^\dagger(\mathbf{p})|0\rangle.$$

For the electron field, the $b^\dagger$ -states are electron states and the $d^\dagger$ -states are positron states. More generally, a Dirac field creates and annihilates a charged fermion and its antiparticle.

A two-fermion state changes sign when the order of the creation operators is exchanged:

$$\hat{b}_s^\dagger(\mathbf{p})\hat{b}_r^\dagger(\mathbf{q})|0\rangle = -\hat{b}_r^\dagger(\mathbf{q})\hat{b}_s^\dagger(\mathbf{p})|0\rangle.$$

This antisymmetry is the defining feature of fermionic Fock space.

4.7 Hamiltonian, Momentum, and Charge

After normal ordering, the free Dirac Hamiltonian is

$$:\hat{H}:= \sum_s \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} \left(\hat{b}_s^\dagger(\mathbf{p})\hat{b}_s(\mathbf{p}) + \hat{d}_s^\dagger(\mathbf{p})\hat{d}_s(\mathbf{p}) \right).$$

The momentum operator is

$$:\hat{\mathbf{P}}:= \sum_s \int \frac{d^3p}{(2\pi)^3} \mathbf{p} \left(\hat{b}_s^\dagger(\mathbf{p})\hat{b}_s(\mathbf{p}) + \hat{d}_s^\dagger(\mathbf{p})\hat{d}_s(\mathbf{p}) \right).$$

Both particles and antiparticles carry positive energy.

For a Dirac field with unit charge convention, the normal-ordered charge has the form

$$\hat{Q} = \sum_s \int \frac{d^3p}{(2\pi)^3} \left(\hat{b}_s^\dagger(\mathbf{p})\hat{b}_s(\mathbf{p}) - \hat{d}_s^\dagger(\mathbf{p})\hat{d}_s(\mathbf{p}) \right).$$

For the electron field one often multiplies this operator by $-e$, so that particles created by $\hat{b}^\dagger$ have charge $-e$, and antiparticles created by $\hat{d}^\dagger$ have charge $+e$.

4.8 Why Anticommutators Give the Correct Energy Spectrum

The negative-frequency part of the Dirac field is not interpreted as a tower of negative-energy particles. After quantization, it is reinterpreted as the creation of antiparticles with positive energy. The anticommutation relations are essential for this reinterpretation to give a Hamiltonian bounded below.

Schematically, the Dirac Hamiltonian before normal ordering contains terms of the form

$$\hat{b}^\dagger \hat{b} - \hat{d} \hat{d}^\dagger.$$

Using anticommutators,

$$\hat{d} \hat{d}^\dagger = 1 - \hat{d}^\dagger \hat{d},$$

so, up to an infinite constant that is removed by normal ordering, the antiparticle contribution becomes positive:

$$\hat{d}^\dagger \hat{d}.$$

This is the algebraic reason the free Dirac Hamiltonian counts both particles and antiparticles with positive energy.

4.9 Spinor Sums and Completeness

The spinor wave functions are normalized so that they satisfy completeness relations. A common convention is

$$\sum_s u_s(\mathbf{p}) \bar{u}_s(\mathbf{p}) = \gamma^\mu p_\mu + m,$$

and

$$\sum_s v_s(\mathbf{p}) \bar{v}_s(\mathbf{p}) = \gamma^\mu p_\mu - m,$$

where $p^0 = E_{\mathbf{p}}$. These identities are used constantly when computing fermion propagators and scattering amplitudes.

The spinor sums play for the Dirac field a role analogous to polarization sums for vector fields. They allow spin labels on internal lines to be summed in a Lorentz-covariant way.

4.10 The Dirac Propagator

The Feynman propagator of the Dirac field is

$$S_F(x - y) = \langle 0 | T \hat{\psi}(x) \hat{\bar{\psi}}(y) | 0 \rangle.$$

In momentum space it is

$$S_F(x - y) = \int \frac{d^4 p}{(2\pi)^4} \frac{i(\gamma^\mu p_\mu + m)}{p^2 - m^2 + i\epsilon} e^{-ip \cdot (x - y)}.$$

Equivalently, the propagator is the Green function of the Dirac operator:

$$(i\gamma^\mu \partial_\mu - m) S_F(x - y) = i\delta^{(4)}(x - y).$$

Compared with the scalar propagator, the denominator is the same mass-shell factor, but the numerator carries spinor structure.

4.11 Chirality, Helicity, and the Massless Limit

The Dirac field contains spin degrees of freedom. For massive fermions, the spin label may be chosen in several equivalent ways. For massless fermions, helicity and chirality become especially important.

The chirality matrix γ^5 defines projection operators

$$P_L = \frac{1}{2}(1 - \gamma^5), \quad P_R = \frac{1}{2}(1 + \gamma^5).$$

They split a Dirac field into left- and right-chiral components:

$$\psi_L = P_L \psi, \quad \psi_R = P_R \psi.$$

For a massless Dirac field, the left- and right-chiral parts decouple. This will be essential later in the electroweak theory, where left- and right-handed fermions transform differently under gauge symmetries.

4.12 Majorana Fields: Conceptual Overview

A Dirac field describes a charged fermion and a distinct antiparticle. A Majorana field is different: it is a fermionic field that is equal to its charge conjugate in an appropriate sense. Informally, a Majorana fermion is its own antiparticle.

The Majorana idea is possible for neutral fermions, not for fields carrying an ordinary conserved electric charge. The full construction requires charge conjugation matrices and reality conditions on spinors. For now, the important contrast is simple:

Dirac field: particle and antiparticle are distinct,

while

Majorana field: particle and antiparticle are identified.

Majorana fields will become relevant when discussing neutrino masses and extensions of the Standard Model.

Summary

The Dirac field is the first fermionic quantum field in the course. Its quantization uses anticommutators rather than commutators. The mode expansion contains particle annihilation operators and antiparticle creation operators, now with spin labels. Anticommutation gives fermionic Fock space, the Pauli exclusion principle, and a Hamiltonian with positive particle and antiparticle energies after normal ordering.

4.13 Chapter Checklist

After reading this chapter, one should be able to:

- write the free Dirac Lagrangian and Dirac equation;
- explain why fermionic fields are quantized with anticommutators;
- write the mode expansion of $\hat{\psi}$ and $\hat{\bar{\psi}}$;
- state the anticommutation relations of $\hat{b}, \hat{b}^\dagger$ and $\hat{d}, \hat{d}^\dagger$;
- construct fermion and antifermion states;
- explain how the Hamiltonian counts positive-energy particles and antiparticles;
- write the charge operator and interpret the sign convention;
- state the standard spinor completeness relations;
- write the Dirac propagator;
- distinguish Dirac and Majorana fermions at the conceptual level.

Chapter 5

Quantization of the Electromagnetic Field

Remark

The scalar and Dirac fields taught us how particles and antiparticles arise from mode expansions. The electromagnetic field adds a new feature: gauge redundancy. A photon is not an excitation of four independent components of A_μ , but of the physical transverse degrees of freedom of a gauge field.

5.1 Why Gauge Fields Are Harder to Quantize

The real scalar, complex scalar, and Dirac fields all had independent field components that could be quantized after imposing the appropriate commutation or anticommutation relations. The electromagnetic field is different. It is described by a vector potential A_μ , but not all components of A_μ represent physical degrees of freedom.

The field strength is

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

The free electromagnetic Lagrangian is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}.$$

It is invariant under the gauge transformation

$$A_\mu \longrightarrow A_\mu + \partial_\mu \Lambda.$$

The potentials A_μ and $A_\mu + \partial_\mu \Lambda$ describe the same physical electromagnetic field. Therefore quantizing every component of A_μ as if it were an independent scalar field would overcount states.

5.2 Physical and Redundant Degrees of Freedom

A massless vector field in four spacetime dimensions has two physical polarizations. This is already suggested classically by electromagnetic waves: the electric and magnetic fields of a plane wave are transverse to the direction of propagation.

The four components of A_μ contain redundant information. Gauge freedom removes one part, and the equations of motion impose a constraint. The result is that only two transverse photon polarizations are physical.

This distinction is central:

$$\text{four components of } A_\mu \neq \text{four physical photon polarizations.}$$

The task of quantization is to keep Lorentz covariance and locality under control while ensuring that only the physical states contribute to observable processes.

5.3 Canonical Obstruction

The canonical momentum conjugate to A_μ is

$$\Pi^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_0 A_\mu)}.$$

For the time component one finds

$$\Pi^0 = 0.$$

This means that A_0 has no independent time derivative in the Lagrangian. It is not an ordinary dynamical coordinate. Instead, it acts as a Lagrange multiplier imposing Gauss's law.

This is the first signal that gauge fields require constraints or gauge fixing. One cannot simply impose four independent canonical commutation relations for all components of A_μ .

5.4 Coulomb Gauge Quantization

A physically transparent gauge choice is Coulomb gauge,

$$\nabla \cdot \mathbf{A} = 0.$$

For the free electromagnetic field, one may also set the nondynamical scalar potential to zero after solving the constraint. The remaining field is the transverse vector potential $\mathbf{A}_T$.

The transverse field may be expanded as

$$\hat{\mathbf{A}}(t, \mathbf{x}) = \sum_{\lambda=1}^2 \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2|\mathbf{k}|}} \left(\epsilon_\lambda(\mathbf{k}) \hat{a}_\lambda(\mathbf{k}) e^{-i|\mathbf{k}|t + i\mathbf{k} \cdot \mathbf{x}} + \epsilon_\lambda^*(\mathbf{k}) \hat{a}_\lambda^\dagger(\mathbf{k}) e^{i|\mathbf{k}|t - i\mathbf{k} \cdot \mathbf{x}} \right).$$

The polarization vectors are transverse:

$$\mathbf{k} \cdot \epsilon_\lambda(\mathbf{k}) = 0.$$

The index $\lambda = 1, 2$ labels the two physical polarizations.

5.5 Photon Creation and Annihilation Operators

The photon ladder operators satisfy

$$[\hat{a}_\lambda(\mathbf{k}), \hat{a}_\sigma^\dagger(\mathbf{p})] = (2\pi)^3 \delta_{\lambda\sigma} \delta^{(3)}(\mathbf{k} - \mathbf{p}),$$

with all other commutators vanishing. The vacuum is defined by

$$\hat{a}_\lambda(\mathbf{k})|0\rangle = 0 \quad \text{for all } \mathbf{k}, \lambda.$$

A one-photon state is

$$|\mathbf{k}, \lambda\rangle = \hat{a}_\lambda^\dagger(\mathbf{k})|0\rangle.$$

Its energy is

$$\omega_{\mathbf{k}} = |\mathbf{k}|,$$

which is the massless dispersion relation.

The normal-ordered Hamiltonian is

$$:\hat{H}:= \sum_{\lambda=1}^2 \int \frac{d^3k}{(2\pi)^3} |\mathbf{k}| \hat{a}_\lambda^\dagger(\mathbf{k}) \hat{a}_\lambda(\mathbf{k}).$$

The momentum operator is

$$:\hat{\mathbf{P}}:= \sum_{\lambda=1}^2 \int \frac{d^3k}{(2\pi)^3} \mathbf{k} \hat{a}_\lambda^\dagger(\mathbf{k}) \hat{a}_\lambda(\mathbf{k}).$$

5.6 Polarization Vectors

For each nonzero momentum $\mathbf{k}$, one chooses two transverse polarization vectors. They obey

$$\mathbf{k} \cdot \epsilon_\lambda(\mathbf{k}) = 0, \quad \epsilon_\lambda^*(\mathbf{k}) \cdot \epsilon_\sigma(\mathbf{k}) = \delta_{\lambda\sigma}.$$

Linear polarizations are useful for many classical comparisons. Circular polarizations are useful for describing helicity. They are built from two linear polarizations by

$$\epsilon_\pm = \frac{1}{\sqrt{2}} (\epsilon_1 \pm i\epsilon_2).$$

The labels $+$ and $-$ correspond to the two helicity states of the photon.

5.7 Photon States and Helicity

A massive spin-one particle would have three spin polarizations. A photon is massless and gauge invariant, so only two physical helicities remain. A one-photon state may be labelled as

$$|\mathbf{k}, +\rangle, \quad |\mathbf{k}, -\rangle.$$

These are helicity eigenstates. Helicity is the projection of angular momentum onto the direction of motion. For a massless particle, helicity is Lorentz invariant under proper orthochronous Lorentz transformations.

The absence of a longitudinal physical photon is not an accident. The longitudinal component of $\mathbf{A}$ is gauge redundancy, not a separate particle.

5.8 Covariant Quantization

Coulomb gauge makes the physical degrees of freedom visible, but it hides manifest Lorentz covariance. For perturbative QFT it is often more convenient to use a covariant gauge-fixing term,

$$\mathcal{L}_{\text{gf}} = -\frac{1}{2\xi} (\partial_\mu A^\mu)^2.$$

The gauge-fixed Lagrangian is

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2\xi} (\partial_\mu A^\mu)^2.$$

The parameter ξ labels a family of gauges. The choice $\xi = 1$ is called Feynman gauge.

Covariant quantization introduces unphysical components into intermediate calculations. These components must not appear in physical observables. In QED, current conservation ensures that unphysical polarizations cancel from scattering amplitudes. In non-Abelian gauge theory, the corresponding statement requires ghosts and BRST symmetry, which will be developed later.

5.9 Gauge Fixing and the Photon Propagator

In covariant gauges, the momentum-space photon propagator is

$$D_{\mu\nu}(k) = \frac{-i}{k^2 + i\epsilon} \left(\eta_{\mu\nu} - (1 - \xi) \frac{k_\mu k_\nu}{k^2 + i\epsilon} \right).$$

In Feynman gauge, this simplifies to

$$D_{\mu\nu}(k) = \frac{-i\eta_{\mu\nu}}{k^2 + i\epsilon}.$$

The pole at $k^2 = 0$ reflects the masslessness of the photon. The numerator carries the vector structure of the field.

The gauge-dependent part proportional to $k_\mu k_\nu$ does not change physical QED amplitudes coupled to conserved currents. This is the practical reason covariant gauges are useful: they simplify calculations without changing observable predictions.

5.10 Unphysical Polarizations

Covariant quantization treats A_μ as a four-component field during intermediate steps. This introduces time-like and longitudinal polarizations that are not physical photon states. The physical Hilbert space is obtained only after imposing the appropriate condition that removes the unphysical sector.

In elementary QED calculations, this removal is often hidden inside Ward identities and current conservation. If an external photon polarization vector is shifted by a multiple of its momentum,

$$\epsilon_\mu(k) \longrightarrow \epsilon_\mu(k) + c k_\mu,$$

physical amplitudes are unchanged. This is the amplitude-level expression of gauge redundancy.

5.11 Gauge-Invariant Observables

The potential A_μ is not gauge invariant. The field strength $F_{\mu\nu}$ is gauge invariant, and so are the electric and magnetic fields. Observables must ultimately be built from gauge-invariant quantities or from gauge-invariant combinations of charged fields and gauge fields.

Examples include

$$F_{\mu\nu}(x), \quad F_{\mu\nu}(x)F^{\mu\nu}(x),$$

and, in more advanced settings, Wilson line and Wilson loop operators. The important lesson at this stage is that the photon is described by a gauge potential, but physical information cannot depend on the redundant gauge choice.

5.12 What Is a Photon?

In free QED, a photon is a one-quantum excitation of a transverse mode of the electromagnetic field:

$$|\mathbf{k}, \lambda\rangle = \hat{a}_\lambda^\dagger(\mathbf{k})|0\rangle, \quad \lambda = 1, 2.$$

Equivalently, one may label the two states by helicity $\lambda = \pm$. The photon is massless, has energy $|\mathbf{k}|$, and carries no electric charge.

This definition is sharp in the free theory and in scattering theory with asymptotic photon states. In interacting QED, soft photons and infrared effects make the exact particle interpretation more subtle. Those subtleties will be revisited when discussing scattering and loop corrections.

Summary

The electromagnetic field is the first gauge field in the course. Gauge redundancy means that not all components of A_μ are physical. In Coulomb gauge the two transverse photon polarizations are explicit. In covariant gauges one keeps Lorentz covariance at the cost of introducing unphysical components, which cancel from physical observables. A photon is a transverse, massless quantum of the electromagnetic field.

5.13 Chapter Checklist

After reading this chapter, one should be able to:

- explain why gauge fields are harder to quantize than scalar fields;
- state the gauge redundancy of A_μ ;
- explain why the photon has two physical polarizations;
- write the Coulomb-gauge mode expansion of the transverse vector potential;
- construct one-photon states;
- distinguish linear polarization from helicity;
- write the covariant gauge-fixing term;
- write the photon propagator in a covariant gauge;
- explain why unphysical polarizations do not appear as external photon states;
- identify gauge-invariant electromagnetic observables.

Chapter 6

Propagators as Quantum Correlation Functions

Remark

The previous chapters constructed free quantum fields and their particle states. This chapter introduces the next central object: the propagator. A propagator is not merely a line in a Feynman diagram. It is a vacuum correlation function of quantum fields, and it encodes both the equation of motion and the particle content of the theory.

6.1 From Classical Green Functions to Quantum Correlators

In classical field theory, a Green function solves an inhomogeneous differential equation. For the Klein–Gordon operator, one writes

$$(\square_x + m^2)G(x - y) = \delta^{(4)}(x - y).$$

Different boundary conditions give different Green functions: retarded, advanced, or other choices.

In quantum field theory, a closely related object appears, but with a different interpretation. The basic two-point function is a vacuum expectation value of a product of field operators. For a real scalar field, the simplest one is

$$\langle 0 | \hat{\phi}(x) \hat{\phi}(y) | 0 \rangle.$$

This is not the value of a classical field at two points. It is a quantum correlation between two field operators in the vacuum.

6.2 Wightman Functions

The positive-frequency Wightman function of the real scalar field is

$$\Delta^+(x - y) = \langle 0 | \hat{\phi}(x) \hat{\phi}(y) | 0 \rangle.$$

Using the mode expansion, one finds

$$\Delta^+(x - y) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_{\mathbf{k}}} e^{-i\omega_{\mathbf{k}}(x^0 - y^0) + i\mathbf{k} \cdot (\mathbf{x} - \mathbf{y})}.$$

The reversed ordering gives

$$\Delta^-(x - y) = \langle 0 | \hat{\phi}(y) \hat{\phi}(x) | 0 \rangle.$$

These functions measure vacuum correlations. Even in the vacuum, field operators at different spacetime points are correlated.

The commutator function is the difference

$$[\hat{\phi}(x), \hat{\phi}(y)] = \Delta^+(x - y) - \Delta^-(x - y) = i\Delta(x - y).$$

Microcausality says that this commutator vanishes for spacelike separation.

6.3 Time-Ordered Products

Scattering theory is built from time-ordered products. For two bosonic field operators, the time-ordered product is defined by

$$T\hat{\phi}(x)\hat{\phi}(y) = \begin{cases} \hat{\phi}(x)\hat{\phi}(y), & x^0 > y^0, \\ \hat{\phi}(y)\hat{\phi}(x), & y^0 > x^0. \end{cases}$$

Equivalently,

$$T\hat{\phi}(x)\hat{\phi}(y) = \theta(x^0 - y^0)\hat{\phi}(x)\hat{\phi}(y) + \theta(y^0 - x^0)\hat{\phi}(y)\hat{\phi}(x).$$

For fermionic fields, time ordering includes a minus sign when two fermion operators are exchanged. This sign is one of the first places where fermionic statistics enters perturbation theory.

6.4 The Feynman Propagator

The scalar Feynman propagator is the vacuum expectation value of the time-ordered two-point function:

$$D_F(x - y) = \langle 0 | T \hat{\phi}(x) \hat{\phi}(y) | 0 \rangle.$$

Using the Wightman functions, this may be written as

$$D_F(x - y) = \theta(x^0 - y^0)\Delta^+(x - y) + \theta(y^0 - x^0)\Delta^+(y - x).$$

It has the four-dimensional momentum-space representation

$$D_F(x - y) = \int \frac{d^4k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} e^{-ik \cdot (x - y)}.$$

The small $i\epsilon$ prescription specifies how the poles are passed in the complex k^0 -plane. This prescription is what makes the propagator the vacuum time-ordered correlation function rather than some other Green function.

6.5 The Propagator as a Green Function

The Feynman propagator satisfies

$$(\square_x + m^2)D_F(x - y) = -i\delta^{(4)}(x - y),$$

with the sign convention following from the definition above. Thus the Feynman propagator is also a Green function of the Klein-Gordon operator, but with a specific quantum boundary condition.

This dual nature is important:

$$\text{propagator} = \text{Green function} = \text{vacuum two-point correlation function}.$$

The first viewpoint emphasizes equations of motion. The second emphasizes quantum states and operator expectation values.

6.6 Retarded and Advanced Propagators

The retarded propagator is built from the commutator:

$$D_R(x - y) = i\theta(x^0 - y^0)[\hat{\phi}(x), \hat{\phi}(y)].$$

It vanishes when $x^0 < y^0$, so it describes causal response to a disturbance in the past. The advanced propagator is

$$D_A(x - y) = -i\theta(y^0 - x^0)[\hat{\phi}(x), \hat{\phi}(y)].$$

It vanishes when $x^0 > y^0$.

Retarded and advanced propagators are especially natural in classical response problems and nonequilibrium physics. The Feynman propagator is the object that appears in ordinary perturbative scattering theory.

6.7 The $i\epsilon$ Prescription

The expression

$$\frac{i}{k^2 - m^2 + i\epsilon}$$

contains more information than the algebraic denominator. The poles are located near

$$k^0 = \pm\omega_{\mathbf{k}}.$$

The $i\epsilon$ prescription tells us how to deform the integration contour around these poles. It is the momentum-space encoding of time ordering.

For positive time separation, the contour picks up the positive-energy pole. For negative time separation, it picks up the negative-energy pole in the way appropriate to a time-ordered vacuum expectation value. This is how the compact four-dimensional expression reproduces the two time orderings in position space.

6.8 Momentum-Space Propagators for Free Fields

For the real scalar field,

$$D_F(k) = \frac{i}{k^2 - m^2 + i\epsilon}.$$

For the complex scalar field,

$$\langle 0 | T \hat{\phi}(x) \hat{\phi}^\dagger(y) | 0 \rangle$$

has the same momentum-space denominator. The difference is that the complex scalar propagator carries charge flow from $\phi^\dagger$ to ϕ .

For the Dirac field,

$$S_F(k) = \frac{i(\gamma^\mu k_\mu + m)}{k^2 - m^2 + i\epsilon}.$$

The denominator is the same mass-shell structure, while the numerator carries spinor information.

For the photon in Feynman gauge,

$$D_{\mu\nu}(k) = \frac{-i\eta_{\mu\nu}}{k^2 + i\epsilon}.$$

In a general covariant gauge, the numerator changes by gauge-dependent terms. Physical amplitudes do not depend on that gauge choice.

6.9 Pole Structure and Particle Interpretation

The location of poles in a propagator reveals the particle content of the free theory. The scalar propagator has poles at

$$k^2 = m^2,$$

which is the relativistic mass shell. This is the analytic trace of a particle of mass m .

For a massless photon, the pole is at

$$k^2 = 0.$$

For a free Dirac field, the denominator again has poles at $k^2 = m^2$, while the numerator encodes spinor structure.

In an interacting theory, the pole structure becomes even more informative. A shift in the pole position corresponds to a mass correction. A pole with an imaginary part corresponds to an unstable resonance. Branch cuts correspond to multiparticle thresholds. These ideas will be developed later, but the free propagator already contains the basic principle: particles appear as singular structures of correlation functions.

6.10 Propagation Versus Correlation

The word “propagator” can be misleading. It suggests that a particle literally travels from y to x . In QFT the primary object is a correlation function of fields. It measures the amplitude associated with inserting one field at y and another at x , with the vacuum in between.

For spacelike separation, the Feynman propagator need not vanish. This does not violate causality, because the commutator vanishes outside the light cone. The causal statement is not that all vacuum correlations vanish at spacelike separation, but that spacelike separated observables commute.

Thus one should distinguish

$$\text{vacuum correlation} \neq \text{signal propagation}.$$

The propagator is a correlation function used in the computation of amplitudes; it is not by itself a directly observable signal.

6.11 What a Propagator Means in Perturbation Theory

In perturbation theory, propagators connect field insertions. Wick’s theorem will show that time-ordered products of free fields can be reduced to sums of contractions, and each contraction is a Feynman propagator.

This is why propagators become internal lines in Feynman diagrams. A diagram is a compact bookkeeping device for a term in the perturbative expansion. The line stands for a two-point function of the corresponding free field.

For example, a scalar internal line carries

$$\frac{i}{k^2 - m^2 + i\epsilon},$$

a fermion internal line carries

$$\frac{i(\gamma^\mu k_\mu + m)}{k^2 - m^2 + i\epsilon},$$

and a photon internal line in Feynman gauge carries

$$\frac{-i\eta_{\mu\nu}}{k^2 + i\epsilon}.$$

The diagrammatic rules are therefore not arbitrary. They are built from the free-field correlation functions developed in this chapter.

6.12 Summary of the Main Propagators

The most important free propagators introduced so far are

$$\begin{array}{ll} \text{real scalar} & : \frac{i}{k^2 - m^2 + i\epsilon}, \\ \text{complex scalar} & : \frac{i}{k^2 - m^2 + i\epsilon} \text{ with charge flow,} \\ \text{Dirac field} & : \frac{i(\gamma^\mu k_\mu + m)}{k^2 - m^2 + i\epsilon}, \\ \text{photon, Feynman gauge} & : \frac{-i\eta_{\mu\nu}}{k^2 + i\epsilon}. \end{array}$$

The denominators encode the mass shell. The numerators encode spin, vector, or gauge structure.

Summary

A propagator is a time-ordered vacuum two-point function. It is also a Green function with a specific quantum boundary condition. Its poles encode particle content, while its numerator encodes spin and gauge structure. Propagators are the building blocks of perturbative QFT because Wick's theorem turns contractions of free fields into propagators.

6.13 Chapter Checklist

After reading this chapter, one should be able to:

- distinguish classical Green functions from quantum two-point functions;
- define Wightman functions;
- define the time-ordered product;
- write the scalar Feynman propagator in position and momentum space;
- explain the role of the $i\epsilon$ prescription;
- distinguish Feynman, retarded, and advanced propagators;
- write the free scalar, Dirac, and photon propagators;
- explain how pole structure encodes particle content;
- explain why vacuum correlation is not the same as signal propagation;
- explain why propagators become internal lines in Feynman diagrams.

Chapter 7

The Interaction Picture

Remark

This chapter is reserved by the course plan in `ideas_Quantum_Field_Theory.md`. It will be developed in small increments.

7.1 Planned Sections

- Schrödinger, Heisenberg, and interaction pictures.
- Free fields as the starting point.
- Interaction Hamiltonian.
- Time evolution operator.
- Dyson series.
- Time-ordered exponential.
- Perturbation theory as an expansion in couplings.
- Vacuum persistence amplitude.
- Adiabatic switching and the meaning of in/out states.
- Why exact interacting QFT is difficult.

Chapter 8

Wick Theorem and Diagrammatics

Remark

This chapter follows the planned QFT table of contents. It is a placeholder to be developed in small increments.

8.1 Planned Sections

- Normal ordering revisited.
- Contractions.
- Wick theorem for scalar fields.
- Wick theorem for fermions.
- Fermionic sign factors.
- From algebra to diagrams.
- External lines, internal lines, and vertices.
- Connected and disconnected diagrams.
- Vacuum bubbles.
- Symmetry factors.
- Interpretation limits of Feynman diagrams.

Chapter 9

Scattering Theory and the S -Matrix

Remark

This chapter is reserved by the course plan in `ideas_Quantum_Field_Theory.md`. It will be developed in small increments.

9.1 Planned Sections

- In-states and out-states.
- The S -matrix.
- Transition amplitudes.
- Matrix elements and momentum conservation.
- Cross sections.
- Decay rates.
- Lorentz-invariant phase space.
- The LSZ reduction formula.
- From correlation functions to scattering amplitudes.
- Tree-level $2 \rightarrow 2$ scattering.
- Decay of an unstable particle.
- From amplitudes to measurable predictions.

Chapter 10

Path Integral for Quantum Mechanics

Remark

This chapter is reserved by the course plan in `ideas_Quantum_Field_Theory.md`. It will be developed in small increments.

10.1 Planned Sections

- Transition amplitudes as sums over histories.
- Time slicing.
- Classical limit and stationary phase.
- Gaussian integrals.
- Generating functionals in ordinary quantum mechanics.
- Sources and correlation functions.
- Why the path integral is useful for QFT.
- Relation to canonical quantization.

Chapter 11

Path Integral for Scalar Fields

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

11.1 Planned Sections

- Functional integration.
- Free scalar generating functional.
- Sources and derivatives.
- Correlation functions.
- Gaussian functional integrals.
- Perturbation theory.
- Diagrammatic expansion.
- Connected functional.
- Effective action.
- Classical field as expectation value.
- One-particle-irreducible functions.

Chapter 12

Fermionic Path Integrals

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

12.1 Planned Sections

- Grassmann variables.
- Berezin integration.
- Gaussian integrals for Grassmann variables.
- Fermionic generating functional.
- Dirac propagator from the path integral.
- Fermion loops and minus signs.
- Determinants from integrating out fermions.
- Majorana fermions in the path integral.
- Physical interpretation and common pitfalls.

Chapter 13

Gauge Fields in the Path Integral

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

13.1 Planned Sections

- Overcounting gauge-equivalent configurations.
- Gauge fixing in the functional integral.
- Faddeev–Popov method.
- Ghost fields.
- Why ghosts are not physical particles.
- Gauge-fixed photon path integral.
- Gauge-fixed Yang–Mills path integral.
- First introduction to BRST symmetry.
- Gauge independence of physical observables.
- Meaning of gauge fixing in quantum theory.

Chapter 14

ϕ^4 Theory as the First Interacting QFT

Remark

Placeholder chapter from the QFT course plan in `ideas_Quantum_Field_Theory.md`.
Detailed lecture text will be added later.

14.1 Planned Role

This chapter will introduce the first interacting scalar quantum field theory and the basic structure of perturbative Feynman rules, loop corrections, counterterms, and renormalized parameters.

Chapter 15

Yukawa Theory

Remark

Placeholder chapter from the QFT course plan in `ideas_Quantum_Field_Theory.md`.
Detailed lecture text will be added later.

15.1 Planned Role

This chapter will study a scalar field coupled to fermions, Yukawa vertices, simple scattering and decay processes, and the low-energy interpretation of scalar exchange.

Chapter 16

Quantum Electrodynamics

Remark

Placeholder chapter from the QFT course plan in `ideas_Quantum_Field_Theory.md`.
Detailed lecture text will be added later.

16.1 Planned Role

This chapter will introduce QED as the quantum theory of charged matter and light, including Feynman rules, spinor chains, Ward identities, and standard tree-level processes.

Chapter 17

Loop Effects in QED

Remark

Placeholder chapter from the QFT course plan in `ideas_Quantum_Field_Theory.md`.
Detailed lecture text will be added later.

17.1 Planned Role

This chapter will cover radiative corrections in QED, including self-energies, vacuum polarization, vertex corrections, charge renormalization, infrared divergences, and inclusive observables.

Chapter 18

Regularization

Remark

Placeholder chapter from the QFT course plan in `ideas_Quantum_Field_Theory.md`.
Detailed lecture text will be added later.

18.1 Planned Role

This chapter will explain why divergent loop integrals appear and how cutoff, Pauli–Villars, dimensional regularization, and subtraction schemes organize intermediate infinities.

Chapter 19

Renormalization

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

19.1 Planned Role

This chapter will develop the standard QFT treatment of renormalized masses, fields, couplings, counterterms, and physical predictions.

Chapter 20

Renormalization Group

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

20.1 Planned Role

This chapter will organize the scale dependence of quantum field theories and introduce running couplings, beta functions, fixed points, and effective descriptions.

Chapter 21

Effective Field Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

21.1 Planned Role

This chapter will introduce effective field theory, separation of scales, matching, power counting, symmetries, operator bases, decoupling, and naturalness.

Chapter 22

Non-Abelian Gauge Theory as a Quantum Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

22.1 Planned Role

This chapter will develop quantum non-Abelian gauge fields, self-interactions, gauge-fixed Feynman rules, ghosts, Ward-type identities, and unitarity checks.

Chapter 23

BRST Symmetry

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

23.1 Planned Role

This chapter will introduce BRST transformations, ghost number, nilpotency, physical states as cohomology, gauge independence, and Slavnov–Taylor identities.

Chapter 24

Quantum Chromodynamics

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

24.1 Planned Role

This chapter will introduce QCD field content, color charge, QCD Feynman rules, gluon self-interactions, asymptotic freedom, confinement, jets, and perturbative versus nonperturbative regimes.

Chapter 25

Symmetries in Quantum Field Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

25.1 Planned Role

This chapter will translate classical symmetries into quantum operators, conserved charges, Ward identities, current correlation functions, selection rules, and constraints on observables.

Chapter 26

Anomalies

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

26.1 Planned Role

This chapter will study symmetries that fail after quantization, chiral currents, triangle diagrams, path-integral measure effects, gauge anomalies, anomaly cancellation, and physical consequences.

Chapter 27

Vacuum Structure and Nonperturbative Effects

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

27.1 Planned Role

This chapter will discuss the quantum vacuum, degenerate vacua, tunneling, instantons, theta vacua, false vacuum decay, semiclassical approximation, and nonperturbative contributions.

Chapter 28

The Quantum Electroweak Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

28.1 Planned Role

This chapter will introduce the quantum electroweak sector, gauge fixing after symmetry breaking, massive vector boson propagators, currents, weak processes, and the low-energy Fermi theory.

Chapter 29

The Standard Model Lagrangian as a Quantum Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

29.1 Planned Role

This chapter will organize the Standard Model field content, representations, gauge interactions, Yukawa sector, Higgs sector, flavor mixing, neutrino masses, anomaly cancellation, and EFT viewpoint.

Chapter 30

Selected Standard Model Processes

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

30.1 Planned Role

This chapter will connect QFT calculations with representative Standard Model predictions and collider observables.

Chapter 31

Euclidean Quantum Field Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

31.1 Planned Role

This chapter will introduce Wick rotation, Euclidean correlation functions, reflection positivity, statistical field theory connections, finite temperature, boundary conditions, and Matsubara frequencies.

Chapter 32

Lattice Field Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

32.1 Planned Role

This chapter will introduce lattice regularization, scalar fields on a lattice, gauge fields on links, Wilson action, Wilson loops, fermion difficulties, Monte Carlo sampling, and the continuum limit.

Chapter 33

Large- N and Semiclassical Methods

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

33.1 Planned Role

This chapter will introduce large- N counting, planar diagrams, saddle-point methods, semiclassical expansion, collective fields, solitons, instanton calculus, and reliability of approximations.

Chapter 34

Conceptual Synthesis I

34.1 Planned Role

Placeholder chapter from the QFT course plan.

Chapter 35

Conceptual Synthesis II

35.1 Planned Role

Placeholder chapter from the QFT course plan.

Chapter 36

Summary: The Architecture of Quantum Field Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

36.1 Planned Role

This chapter will summarize quantization, Fock space, propagators, interactions, renormalization, gauge theory, anomalies, nonperturbative effects, the Standard Model, and open directions.

Appendix A

Quantization Dictionary

Remark

This appendix records the translation rules that are used repeatedly in the main text. They should not be treated as mechanical replacements valid in every context. They are starting points that must be checked against constraints, symmetries, domains of operators, and gauge redundancies.

A.1 Classical Mode to Quantum Oscillator

A classical normal mode with amplitude $q_{\mathbf{k}}(t)$ and frequency $\omega_{\mathbf{k}}$ is quantized like a harmonic oscillator. Schematically,

$$q_{\mathbf{k}}, p_{\mathbf{k}} \longrightarrow \hat{q}_{\mathbf{k}}, \hat{p}_{\mathbf{k}}, \quad [\hat{q}_{\mathbf{k}}, \hat{p}_{\mathbf{k}'}] = i\delta_{\mathbf{k}\mathbf{k}'}.$$

The oscillator is then rewritten in terms of ladder operators $\hat{a}_{\mathbf{k}}$ and $\hat{a}_{\mathbf{k}}^\dagger$.

A.2 Poisson Bracket to Commutator

The canonical rule is

$$\{A, B\}_{\text{PB}} \longrightarrow \frac{1}{i}[\hat{A}, \hat{B}].$$

For fields, the equal-time canonical relation is

$$[\hat{\phi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{y})] = i\delta^{(3)}(\mathbf{x} - \mathbf{y}).$$

A.3 Classical Field to Field Operator

A classical field $\phi(x)$ becomes an operator-valued distribution $\hat{\phi}(x)$. The word “distribution” is important: products of field operators at the same spacetime point are generally singular and require a careful definition.

A.4 Classical Green’s Function to Propagator

A classical Green’s function solves a differential equation with a delta-function source. A quantum propagator is a vacuum expectation value of a time-ordered product,

$$\Delta_F(x - y) = \langle 0|T\{\hat{\phi}(x)\hat{\phi}(y)\}|0\rangle.$$

It is related to Green’s functions, but its physical meaning is quantum correlation, not merely classical signal propagation.

A.5 Classical Symmetry to Quantum Ward Identity

A classical continuous symmetry gives a conserved current. In the quantum theory, the same symmetry constrains correlation functions and scattering amplitudes through Ward identities. If the symmetry fails after quantization, the failure is called an anomaly.

Appendix B

Reference B

Placeholder appendix.

Appendix C

Reference C

Placeholder appendix.

Appendix D

Reference D

Placeholder appendix.

Appendix E

Reference E

Placeholder appendix.

List of Figures